



Chaitanya Priya

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SKILLS

Embedded C | C | C,C,C++ | Protocol: CAN, CAN FD, UDS, I2C, UART | FreeRTOS | MISRA | SPI I2C | OpenOCD and JTAG debugging | Microcontrollers-16/32bit | IBM-RTC | Polyspace BF & CP | printed circuit boards | KiCAD , Altium CAD tools | Lauterbach Trace 32 | GIT (GitHub) | Semaphores | Reading Schematics | Electronic Circuits | IBM DOORS NG | ISO standards: ISO 21434, ISO 26262 | Vector tools (CANoe, DaVinci Configurator) | ETAS RTA BSW | ETAS RTA OS | Mathworks MATLAB & Simulink | VectorCast(Unit test) | Vector CANoe | Functional safety ISO26262 | SAE J1939 | Python | UDS ISO14229 | Vector vTESTstudio | Management tools (JIRA, Confluence) | Autosar MCAL and BSW

WORK EXPERIENCE

SENIOR TECHNICAL LEAD – MERCEDES-BENZ GMBH FROM SOORCE GMBH – 01/10/2024 – Current – STUTTGART, GERMANY

- Developed and maintained the Power State Manager (PSM) software module, orchestrating the critical power-on sequence for Aurix TCx-based boards and multi-OS System-on-Chips (SoC).
- Engineered multi-stage logic to sequence hardware initialization, transitioning control from the Aurix TCx-based boards to the SoC (QNX) and finally to Linux via an embedded Hypervisor.
- Architected an automated Jira triage pipeline using Retrieval-Augmented Generation (RAG) to classify incoming issues, resulting in the automated rejection and redirection of out-of-scope tickets to Project Management.
- Implemented AI-driven root cause analysis within the triage pipeline to identify malfunctioning software modules and generate targeted code edit suggestions, accelerating the bug resolution lifecycle.
- Collaborated with the Architecture Team to define and refine the "OnOff" system architecture, ensuring robust power states and high-availability startup sequences for safety-critical automotive systems.
- Resolved complex software defects by owning the end-to-end lifecycle of incoming bug tickets, from initial technical analysis on the hardware-software interface to final implementation.

SENIOR TECHNICAL LEAD – VECTOR INFORMATIK GMBH BY MSK ENGINEERING AND IT SOLUTIONS GMBH – 17/07/2023 – 09/2024 – STUTTGART

- Led requirement elicitation and developed comprehensive design documentation for EV charging sequences.
- Engineered complex device drivers and designed robust low-level software architectures.
- Designed software solutions and seamlessly integrated third-party libraries into existing architectural frameworks.
- Configured and maintained AUTOSAR base software, ensuring optimal system performance.
- Coordinated globally with customers and vendors to facilitate the integration of third-party libraries. Developed rigorous test and safety plans, ensuring full compliance with ISO 26262 standards.
- As Feature Owner, executed comprehensive Unit, static code analysis, SIL, and integration testing for assigned components.
- Represented the team in ASPICE Quality and Cybersecurity Audits (ISO 21434), ensuring compliance and successful outcomes.

LEAD EMBEDDED SOFTWARE ENGINEER – AMERICAN AXLE AND MANUFACTURING. (WWW.AAM.COM) – 01/10/2019 – 17/07/2023 – DETROIT, UNITED STATES

- As Product owner of Motor control electronic control unit, develop device drivers using embedded C for RTA-OS integration.
- Creating a robust test plan and perform SIL /HIL Tests to reduce bugs in production.
- Embedded C/C++ software development for Diagnostic as per ISO14229.
- Integration of third-party libraries e.g Infineon SafeTlib, Cycliclib
- AUTOSAR Configuration of Base software and Microcontroller abstraction layer.
- Unit testing, static code analysis, SIL test, and integration test for component as feature owner.
- Board bring-up, driver development, and BSP configuration for POC and new platform project.
- Coached colleagues for blocking issues and helped achieve their delivery milestones.

- Device driver development of SPI and I2C-based CAN and external memory modules.
- Migration of application layer feature from 16 bit microcontroller to 32 bit Renesas platform.
- Test developed feature by Model in Loop (MIL) and Software in loop (SIL).
- Generation and Integration of C/C++ as Embedded coder.
- Maintainance and development of CI/CD pipeline for smooth vehicle payload deliveris.
- Integrate build debug test code to JDOS (John Deere RTOS) based vehicle project.
- Worked on defect resolution and root cause analysis of defects using 8D, FMEA, and DFMEA techniques.
- Conducted reviews of specification, Code, and quality reports and recommended necessary action.
- Evaluate technical feasibility and commercial aspects of change requests in collaboration with the offshore project management and development team.

EMBEDDED SYSTEMS SOFTWARE DEVELOPERS – LIBRATHERM INSTRUMENTS PVT. LTD. – 01/02/2015 – 01/09/2017 – MUMBAI, INDIA

- Developed and maintained firmware for a range of medical devices, ensuring compliance with FDA regulatory standards.
- Programmed applications and modules for STM32 microcontrollers using C/C++.
- Integrated communication technologies and IoT protocols (MQTT, ModBus RTU/ASCII) into device firmware.
- Created comprehensive documentation for product approval processes.
- Led the product lifecycle from initial concept through to mass production and aftersales support.
- Conducted unit tests to ensure firmware reliability and performance.

EDUCATION AND TRAINING

CERTIFICATE IN GENERATIVE AI FOR PROFESSIONAL Indian Institute of Technology

Website <https://certifications.iitpatna.com/> | Field of study AI ML Deep Learning

ISO21434:2021 CERTIFIED AUTOMOTIVE CYBERSECURITY PRACTITIONER TÜV SÜD

Website <https://www.tuvsud.com/> | Field of study Cyber Security

ISO26262:2018 LEVEL1 AUTOMOTIVE FUNCTIONAL SAFETY ENGINEER PROFESSIONAL CERTIFICATION Omnex Inc.

Website <https://www.omnex.com/> | Field of study ISO 26262

EMACHINES FOR ELECTRIC VEHICLE APPLICATION Devise Electronics Pvt. Ltd

Website <https://www.deviseelectronics.com/> | Field of study E-Motor control technique and HW

Pune, India

CERTIFICATE IN EMBEDDED SYSTEM DESIGN University of Pune

Website <http://www.unipune.ac.in/> | Field of study Embedded Systems Engineering

B.SC (ENGINEERING) Magadh University

Website <https://www.magadhuniversity.ac.in/> | Final grade Distinction

LANGUAGE SKILLS

Mother tongue(s): **HINDI**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C2	C2	C2

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
GERMAN	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user